



Syahrul Fathoni Ahmad

Bachelor of Engineering
Computer Engineering
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syahrul-portfolio-web

Education

Degree	Institute	Current CGPA	Year
S.T., Teknik Komputer	Institut Teknologi Sepuluh Nopember, Surabaya	3.43/4.0	Aug 2025

Skills

- **Full-Stack Development:** Laravel, Python, React.js, Next.js, Express.js, Go (Golang), Flask, API, SQL, MongoDB, Three.js, HTML/CSS
- **AI & IoT Systems:** PyTorch, TensorRT, YOLO, LSTM, NLP Jetson Nano, ESP32, Arduino, C++, PyQt, RTMP/RTSP Streaming
- **DevOps & Cloud:** Git, Docker, Nginx, Vercel, Azure, Apache

Work Experiences

- **PT. Aldzama** Oct 2025 - Present
Software Engineer Gresik, Indonesia
 - **HRIS & ERP Development (Full-Stack):**
 - ◊ Developed and maintained an **on-premise HRIS** supporting **~600 employees**, managing the full lifecycle (frontend, backend, deployment) for attendance, timesheets, and leave requests.
 - ◊ Built a **payroll integration** module with automated Excel exports for direct bank transfers.
 - **DevOps & Code Management:**
 - ◊ Managed **GitHub version control** and deployment pipelines for **13 active projects**.
 - ◊ Conducted **Pull Request** reviews and code merging to maintain codebase quality prior to internal and public releases.
 - **Company Profile & CMS Integration (Full-Stack):**
 - ◊ Designed and developed the frontend and backend for **two CMS-integrated corporate websites**, successfully launching the primary site and currently building the subsidiary platform.
 - **Project Management:**
 - ◊ Led development of the **Refractory Maintenance Dashboard** as **Project Manager**, coordinating technical teams to align with business requirements and timelines.
 - **IT Infrastructure & Server Administration:**
 - ◊ Configured a **secondary cloud storage** backup for the primary Synology server to ensure data security and redundancy.
 - ◊ Implemented a lightweight **internal mail server** to scale company email capacity with zero additional third-party costs.
 - ◊ Administered **on-premise servers** and configured **VPNs** to ensure secure internal access and optimize overall network performance.
- **Freelance** 2022 - 2025
Freelancer Hybrid
 - **JAREE ITS (Academic Journal Platform):** July 2025 - Oct 2025
 - ◊ Orchestrated a seamless, live migration of the OJS platform (v2 to v3), ensuring service continuity for **30k+ users** while upgrading a database of **597k+ records**.
 - ◊ Engineered the complete academic peer-review workflow and developed a **custom front-end theme** to enhance user experience.
 - **Retrux-Shelf-Eye (AI Inventory Management System):** Aug 2025 - Sep 2025
 - ◊ Developed a real-time shelf stock monitoring application using Python, with a user interface built in **PyQt**.

- ◊ Implemented an advanced object detection model (**RT-DETR**) using **SAHI** for sliced inference.
- ◊ Engineered key software utilities based on client needs, including **multi-camera support** and dynamic directory management.

◦ Padel-Clipper (IoT Video Highlight System):

Aug 2025

Tools: Python, Flask, ESP32, Express.js, PHP

- ◊ Developed the **Flask**-based monitoring dashboard and integrated it with a team-built **Express.js** backend for a real-time video clipping service.
- ◊ Engineered and programmed an **IoT push-button device** using **ESP32** and a custom PCB to trigger highlight captures from IP cameras.
- ◊ Configured and deployed the entire system on a **VPS**, successfully scaling the product to serve nearly **250 active customers**.

◦ 3D Container Loading Optimizer:

July 2025 - Aug 2025

Tools: Python, Golang, FastAPI, React.js, Three.js

- ◊ Developed a web-based logistics tool to solve the 3D Bin Packing Problem, achieving a container fill rate of **87-89%**.
- ◊ Researched and implemented multiple packing algorithms in Python, including **Genetic Algorithm**, BLF, and CLPTAC method to find the optimal solution.
- ◊ Built the full-stack application with **React.js**, **FastAPI**, **Golang** and a **Three.js** canvas for 3D visualization.

• PT. Telkom Indonesia Regional V

Feb 2024 - Apr 2024

Internship Programmer (ROC Department)

Surabaya, Indonesia

◦ Automation & Data Pipeline Development:

- ◊ Reduced data retrieval latency by over **80%** by engineering an automated web scraping pipeline with Python and Apache Airflow.
- ◊ Developed and deployed 2 OCR automation with validation step.

◦ System Integration & Reliability:

- ◊ Developed a resilient VPN auto-reconnect system (Power Automate) that effectively **eliminated connectivity downtime** and prevented work disruptions.
- ◊ Created a suite of 7 integrated tools that reduced manual data processing tasks by **40%**.

• SDN Klampis Ngasem I

Aug 2022 - Jul 2023

Robotics Extracurricular Tutor

Surabaya, Indonesia

- ◊ Developed and assembled **Arduino-based Obstacle Avoidance Robots**, preparing **10+ units** for student use and aligning lessons with an existing robotics curriculum to support progressive learning.
- ◊ Mentored students in the **BARONAS competition** by ITS Electrical Engineering, collaborating with a team of three instructors to deliver hands-on sessions, leading the team to reach the *semi-finals*.

Individual Projects

• Smart Finance App (Personal Finance Management System)

Sep 2025

Tools: Go (Golang), React.js, Python, Scikit-learn, PostgreSQL



- ◊ Developed a full-stack personal finance management application that helps users track income and expenses with intelligent categorization. The system integrates a Go backend, React frontend, and a Python AI service to provide automatic transaction categorization using machine learning.

• Padel-Clipper (IoT Video Highlight System with Golang)

June 2025 - August 2025

Tools: Go (Golang), RESTful API, Flask, Python, PostgreSQL



- ◊ Developed a RESTful API in Go (Golang), leveraging goroutines and channels to efficiently handle concurrent requests from multiple IoT devices for a real-time video clipping service.

• Vehicle Speed Estimation System Using Drone with Jetson Nano-based YOLOv8 Method

Feb 2025 - May 2025

Tools: Python, YOLOv8, Jetson-Nano, TensorRT, RTMP



- ◊ Developed a real-time vehicle speed estimation system, achieving a peak measurement accuracy of **99.55%** and consistent performance of over 90% at an optimal 20-meter altitude.
- ◊ Trained a superior YOLOv8 model for night-time conditions (mAP50 of **0.93**) and leveraged **TensorRT** to triple inference speed (from **>90ms** to **<35ms**), enabling real-time processing on the Jetson Nano.

- **BengkelMate: Workshop Business Process Digitalization** Oct 2024 - Jan 2025
Tools: MERN Stack (MongoDB, Express.js, React.js, Node.js) [\[G\]](#)
 - As part of a three-person team, developed a web application that digitized the workshop workflow through a role-based system for **Security, Service Advisor, and Spare Parts teams**.
 - Delivered a high-quality user experience, validated by excellent Lighthouse scores in **Performance (96)**, **Best Practices (96)**, and **SEO (92)**, ensuring a fast and reliable tool for daily operations.

Campus Organization Experiences

- **BK MWA-WM ITS** Jul 2023 - Jul 2024
Head of Human Resources Surabaya, Indonesia
 - Managed the end-to-end employee lifecycle and cultural development for **40+ staff**, encompassing recruitment (**processing 50+ applicants**), training, and employee well-being initiatives.
- **TEKMEET 2023** Jan 2023 - Mar 2023
Project Officer Surabaya, Indonesia
 - Organized sports tournament for **300+ students** and coordinated charity event serving **50+ people**.

Certifications

- **Intern – PT. Telkom Indonesia Regional V**, [Internship-Certificate](#) Feb 2024 - Apr 2024
- **Intermediate Machine Learning by Kaggle**, [Course](#) Dec 2023
- **Feature Engineering by Kaggle**, [Course](#) Dec 2023
- **Marketing Content with Gapura Digital Google**, [Course](#) Sept 2021